

1. (a) What set is the union of $\{8, 16, 22\}$ and the intersection of $\{4, 5, 6, 14, 15, 16, 24, 25, 26\}$ and $\{32, 24, 16, 8\}$?

Answer:

$$\begin{aligned} & \{8, 16, 22\} \cup (\{4, 5, 6, 14, 15, 16, 24, 25, 26\} \cap \{32, 24, 16, 8\}) \\ &= \{8, 16, 22\} \cup \{16, 24\} \\ &= \boxed{\{8, 16, 22, 24\}} \end{aligned}$$

- (b) What set is the intersection of $\{10, 30, 20\}$ and the union of $\{10, 20\}$ and $\{20, 40\}$?

$$\begin{aligned} & \{10, 30, 20\} \cap (\{10, 20\} \cup \{20, 40\}) \\ &= \{10, 30, 20\} \cap \{10, 20, 40\} \\ &= \boxed{\{10, 20\}} \end{aligned}$$

- (c) What is $\{3, 5, 7, 9\} - \{7, 6, 5, 4\}$?

$$\boxed{\{3, 9\}}$$

- (d) What is $\{4, 6, 5, 7\} - \{3, 5, 7, 9\}$?

$$\boxed{\{4, 6\}}$$

- (e) Let $A = \{100, 200, 4, 2, -8\}$. Let $B = \{0, 4, -8, 100, 200\}$.
What is $(A - B) - (B - A)$?

$$\begin{aligned} (A - B) &= \{2\} \\ (B - A) &= \{0\} \\ \{2\} - \{0\} &= \boxed{\{2\}} \end{aligned}$$

- (f) Let $A = \{100, 200, 4, 2, -8\}$. Let $B = \{0, 4, -8, 100, 200\}$.
What is $A - (A - B)$?

$$\begin{aligned} (A - B) &= \{2\} \\ A - \{2\} &= \boxed{\{100, 200, 4, -8\}} \end{aligned}$$

- (g) Use set builder notation to express $\{3, 7, 11, 15, \dots, 403\}$, as compactly as possible, with as little English as possible.

Assuming $0 \in \mathbb{N}$

$$\boxed{\{x \in \mathbb{N} \mid 3 + 4x \leq 403\}}$$

2. Consider the following sets: $H = \{\text{herbivores}\}$. $S = \{\text{animals with sharp teeth}\}$. $R = \{\text{animals that run fast}\}$. $L = \{\text{animals with long necks}\}$.

- (a) Use set notation (e.g., union, intersection, etc) to describe herbivores that have sharp teeth or run fast. Your expression should be as simple as possible. *Do not use set builder notation.*

$$H \cap (S \cup R)$$

- (b) To survive the next ice age, members of a species must have sharp teeth or run fast, but they must also have long necks or have sharp teeth. Express these conditions using set notation, minimizing redundancy.

$$S \cup (L \cap R)$$

- (c) In a certain herbivore ecosystem, there are 52 species that have sharp teeth or can run fast. 18 of these species have both qualities. 25 species have sharp teeth. How many of the 52 species can run fast? (Briefly show how you calculated this, and name the most relevant concept in the calculation)

It is ok to use the symbol of a set to denote its size, if context is clear; no need to use $|$.

Based on the inclusion-exclusion principle that states $S + R = (S \cup R) + (S \cap R)$

$$R = (S \cup R) + (S \cap R) - S$$

$$= 52 + 18 - 25$$

$$= \boxed{R = 45}$$

(Help recived from Rahul Rao)

- (d) The ecosystem mentioned in (c) contains 100 species.

- i. How many don't have sharp teeth and can't run fast? Use deMorgan's laws.

deMorgan's Laws state:

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

$$\therefore \overline{S \cup R} = \overline{S} \cap \overline{R} = 48$$

- ii. How many don't have sharp teeth or can't run fast (or both)? Use deMorgan.

$$\text{As deMorgan's laws state } \overline{\overline{S \cup R}} = \overline{\overline{S} \cap \overline{R}}$$

3. 100 people participated in a brief poll. Everyone answered either yes or no to each question. The first question was “Do you like cookies?”, to which 45 answered yes. The second question was “Do you like milk?”, to which 21 answered yes. You observe that 15 people liked both cookies and milk. Draw a Venn diagram that represents the sets of people who like cookies, and those who like milk (among those who took the poll). Place numbers in your figure so that everyone is represented. Each disjoint “section” of the diagram should have a number that corresponds only to that section. By definition “sections” do not overlap.
4. Let $f(n) = n^2 - 10n + 27$. Prove that $f(9) < f(11)$. This must be done by writing the expression for $f(9)$ on the first line, then on each new line starting with the symbol for equality or inequality, and making one algebraic step. The last line should conclude with $f(11)$, so that when we read all of the lines together we see a chain of equalities or inequalities, such that the first and last expression show what we want to prove. This exercise is more about proof style than algebra. You should aim for simplicity, efficiency, and clarity.

Proof: $f(9) < f(11)$

Let $f(n) := n^2 - 10n + 27$ where n is a real number

$$f(9) = 9^2 - 10(9) + 27 = 18$$

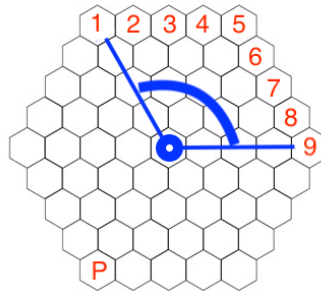
$$< (9 + 2)^2 - 10(9 + 2) + 27$$

$$< 38 = f(11)$$

$$\therefore f(9) = 18 < f(11) = 38$$



5. Suppose that we play a variant of *Chomp* on a hexagonal board like the one below. The poisoned cookie is marked as **P**. Whenever you take a cookie, you must also take everything within its “upper-right” 120° cone. For example, if you take the middle cookie in the illustration then you must take cookies 1 through 9 and everything else in the marked region.



Prove that Player 1 can win this game, with a non-constructive proof.